

### Instructions.

- This is an individual assignment. No collaboration is allowed.
- Missing the deadline will cost you 100% marks.
- Submission will be only on MSTeams.
- This assignment makes 1% of your overall grade.
- Submit one PDF file. It should be named u2023XXX\_a01.pdf where XXX are the last 3 digits of your registration number.

### Question (Max marks 40 – 10 each).

This is a warmup assignment. Do the following tasks:

**Task 1:** Setup a GPU coding environment for yourself. It should give you access to:

- A GPU instance, with
- NVIDIA CUDA installed with appropriate compiler, debugger, etc., and
- Running a CUDA compiled program on the GPU.
- Create github repo for your project and share the link with instructor.
- Push all the detailed description of this task on your repo in the file assignment01/task01.md. Format properly.

**Task 2:** Write a CPU program. It should add two matrices element wise. The input data will be provided in a file. You can define the file structure yourself and explain in you submission. The program when run will take max two filenames as argument. The first is the inputdata filename and the second (optional) argument is the output filename; if the output filename is absent it should write its output to stdout. The program will read input data (matrices) from the input file, store them in CPU memory, perform computations on them, write the result to output file in the same format structure you defined for input matrices. Compile run the code and push it to your github repo in the folder assignment01/task02. Use a build system i.e., Make, CMake, etc., for compilation. Write modular and parametrizable code. Activate compiler options which flag all warnings and report them. You program should compile with 0 warnings and 0 errors.

**Task 3:** Port the above program to GPU/CUDA. The program will read input data (matrices) from the input file, store them in CPU memory, copy them to GPU memory, perform operations on them, copy the result to CPU memory, and write it to output file in the same format structure you defined for input matrices. Compile run the code and push it to your github repo in the folder assignment01/task03. Use a build system i.e., Make, CMake, etc., for compilation. Write modular and parametrizable code. Activate compiler options which flag all warnings and report them. You program should compile with 0 warnings and 0 errors.

**Task 4:** Add code for measuring the computation time to both above programs. GPU time should include data transfer (to and from GPU) as well as computation time. Run both above programs for different matrix sizes and produce a graph of matrix size (x-axis) vs time (y-axis). Comment on what you see. Push it to your github repo in the folder assignment01/task04.

**The End.**